

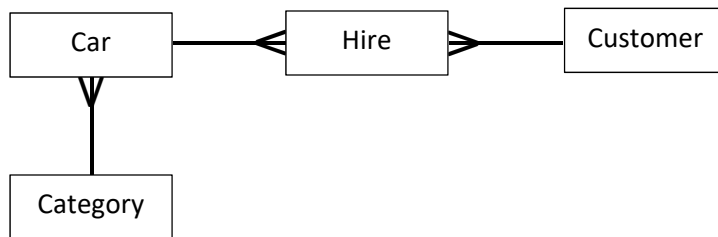
1(a) (i) The table CAR is not in third normal form. Transitive dependency exists. DayRate is transitive dependent on RegistrationNumber, via Category.

(ii) There will be data redundancy as cars with the same category will have the same DayRate value stored in the table. This will lead to data inconsistency if an error is made during insert, update or delete operation.

1(b)

Table	Primary Key	Foreign Key
Car	RegistrationNumber	CategoryName
Category	CategoryName	
Customer	DriverLicenceNumber	
Hire	RegistrationNumber, DriverLicenceNumber, DateHired	DriverLicenceNumber RegistrationNumber

1(c)



1(d) (i)

Two ways the driver licence number can be verified on entry:

- Double data entry - driver licence number to be entered twice.
- Proofreading – have someone checking the driver licence enter against the original document.

(ii)

Three ways the date expected back can be validated on entry:

- Presence check – ensure date expected back is not blank
- Range check – ensure cars are hired for a minimum period of one day
- Format check – correct date format is entered

1(e)

2 advantages to the company of creating a native application instead of a web application

- Able to access device/machine features – application has access to the hardware and software features. This allow the application to be more feature-rich.

- Improved performance – native application offer better performance in terms of speed, responsiveness. The application loads faster, run smoother and deliver better user experiences.

2 disadvantages

- Development for native applications would lead to higher costs because the application had to be developed for multiple platforms.
- Native applications require installation and updates on each individual device.

2(a) Two benefits of using static memory allocation for the data structure.

- There is no need to allocate and deallocate memory during program execution which leads to faster access time.
- Reduce complexity of implementation of data structure as there are no need to manage memory dynamically, this would prevent memory leaks.

(b) Two benefits of using dynamic memory allocation for the data structure

- Allows data structures to adjust their size during runtime which enables efficient utilization of memory resources.
- Optimize memory usage by allocating memory only when needed thus reduce memory wastage compared to static allocation, where memory is allocated regardless of whether they are to be used.

(c) One example where static memory allocation is more appropriate then dynamic memory allocation.

Small datasets such as list of country codes can be statically allocated as their size is known and small and it does not change frequently.

(d) One example of a data structure where dynamic memory allocation is more appropriate than static memory allocation.

Binary trees.

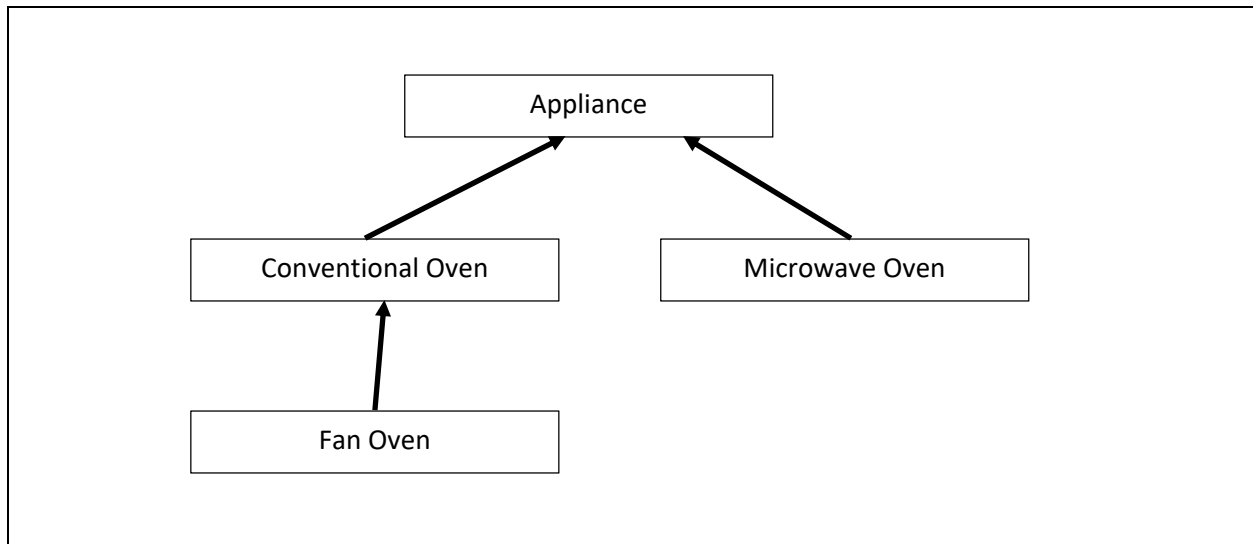
Size of a tree is not known in advance and can change dynamically depending on the number of elements added and deleted during execution hence dynamic memory allocation allows nodes to be allocated and deallocated as needed, accommodating the changing size of the tree.

```
FUNCTION find( list: ARRAY, value: INTEGER, head: INTEGER) RETURNS
INTEGER
    current <- head
    WHILE current < > - 1 DO
        IF list[current].data = value
        THEN
```

```
        RETURN current
    ELSE
        current <- list[current].next
    ENDIF
ENDWHILE
RETURN -1
ENDFUNCTION
```

```
FUNCTION push(stack: ARRAY, value: INTEGER, top: INTEGER) RETURNS
BOOLEAN
    IF top < 100
    THEN
        top <- top + 1
        stack[top] <- value
        RETURN True
    ELSE
        RETURN False
    ENDIF
ENDFUNCTION
```

3(a)



(b) Attributes:

- switch
- timer
- alarm

Methods:

- turnSwitchOn()
- turnSwitchOff()
- setTimer()
- startTimer()
- endTimer()
- startAlarm()
- stopAlarm()

(c) Some attributes of the classes should be declared as private to prevent direct access to them from outside the class. This ensure that only the class method has access to the attributes. Ensure that the class's internal state remains consistent and prevents external code from modifying or corrupting the object's attribute data.

Methods of the class are declared private as they are internal operations that are not relevant or meaningful to external users of the class.

Polymorphism allows methods in the subclass to have the same name as in the superclass to allow for different behaviour.

OR

Polymorphism is the object's ability to **take different forms**. It allows the methods in the subclass to have the same name as the superclass but with different behavior.

4(a) The advantage of storing ordered data in an array instead of a linked list:

- able to traverse data in ascending and descending order using indices of the array
- able to implement a binary search algorithm when searching for data in the array.

(b)

Start with an empty binary search tree where there are no nodes. Inserting the first data item into the binary search tree, it becomes the root of the tree.

For each of the subsequent data item, repeat the following:

Compare the data item with the existing node starting from the root.

Comparison process:

If the value of the data item is less than the value of the current node, go to the left subtree

If the value of the data item is more than the value of the current node, go to the right subtree

The comparison process continues at each subtree repeatedly until an empty spot is found and the new data item will be inserted as a leaf node.

(c) The shape of the binary search tree may differ when the order of inserting of the data items changes.

(d)

(i) Two operations that can performed on a queue : enqueue and dequeue

(ii) Two pointers that are required to implement a queue:

- Front pointer pointing to the start of the queue
- Rear pointer pointing to the end of the queue

(iii) One application of a queue:

- Print job queue which holds the print jobs in a printer server
- Music song playlist
- Job scheduling where tasks or processes are added to the queue and executed in the order they are received.

(iv) The advantage of implementing circular queue instead of a linear queue:

- More efficient use of memory. In a linear queue, once the queue becomes full and items are dequeued, there will be unused space at the beginning of the queue. In a circular queue, the front and rear pointers wrap around the array, allowing the efficient utilization of memory without wasted space.

5(a) (i) The network application creates a digital signature in the following steps:

- Use a hash algorithm to create a hash value of the message to be signed.
- The hash value is encrypted using the private key of the signer.
- The encrypted hash value, the signing algorithm and identity of the signer forms the digital signature.

(ii) The digital signature is authenticated in the following steps:

- Using the signer's public key, the recipient decrypts the encrypted hash value contained within the digital signature to obtain the original hash value of the message.
- The recipient independently compute the hash value of the received message using the same hashing algorithm that was used by the signer to create the digital signature.
- The decrypted hash value obtained from the digital signature (step 1) is compared with the hash value computed from the received message (step 2). If the two hash values match, it indicates that the message has not been tampered with and that the digital signature is authentic.

(b)(i) The format of IPV4 address is represented as a sequence of 4 numbers separated by periods. Each number represents one byte (8bits) of the IP address.

OR

IPV4 address is 32 bits or 4 groups of denary numbers between 0 and 255, separated by dots when written in dotted-decimal notation.

(ii) Two ways that a host can be allocated an IP address:

- Static IP address assignment- administrator manually configures the IP address setting for each host.
- Dynamic IP address allocation – Host requests an IP address by sending a Dynamic Host Configuration Protocol (DHCP) request to a DHCP server in the network which the host is connected to. The DHCP server then dynamically assigns an available IP address from a predefined pool to the host.

(iii) A packet-switching network is a type of data communication network in which data is broken down into smaller units called packets for transmission across the network. The packets are routed independently using different routes through the network from the source to the destination.

(iv) Router facilitates the transmission of data packets from the source to the destination. It forwards data packets from one network segment to another by making routing decision based on its routing table which contains information about the available network paths and the next hop for each data packet based on its destination address.

(c) (i) Denary base-10 value of hexadecimal 4D: 77

(ii) Binary representation of hexadecimal 4D: 0101 1101

(iii) One advantage of using UTF-8 encoding rather than using ASCII is UTF-8 supports a wider range of characters compared to ASCII. UTF-8 can represent over a million characters which includes characters from various languages, symbols, emoji and special characters.

6(a)

The Quicksort algorithm selects a pivot element from the array. It then partition the array into two subarrays: elements less than the pivot and elements more than the pivot. This partitioning is performed such that all elements smaller than the pivot are placed to the left of the pivot and all elements greater than the pivot are placed to the right.

After partitioning, the Quicksort algorithm recursively applies the same process to the subarray of elements less than the pivot and the subarray of elements greater than the pivot.

The recursion continues until each subarray contains zero or one element (base case of the recursive process)

(b)Worst-case time complexity of Quicksort: $O(n^2)$.

(c) (i) Recursion is a programming technique in which a function calls itself to solve a problem by breaking it down into smaller, similar subproblems, until a base case is reached.

(ii)

```
FUNCTION recursive(number1)
    total = 10
    IF number1 = 0 THEN
        RETURN total
    ELSE
        RETURN recursive(number1 - 1) + 3
    ENDIF
ENDFUNCTION
```

7.

The pharmacy must ensure that the customers has given their consent to collect, use or disclose personal data. The pharmacy should allow customers to withdraw consent, with reasonable notice, and inform them of the likely consequences of withdrawal. Upon withdrawal, and depending on the withdrawal request, the pharmacy must cease to collect, use or disclose the customers' personal data.

The pharmacy must ensure that the purpose of data collection is clearly communicated to the customers and only collect data that is necessary for the stated purpose.

The pharmacy must ensure that upon request from the customers, the personal data of the customers and information about the ways in which the data may be used or disclosed in the past year should be provided. The pharmacy must provide the customer with the right to access and correct their personal data, establish procedures for handling such requests in a timely manner.

The pharmacy must develop a data retention policy that outlines how long personal data will be retained and the criteria for its disposal. Personal data should be disposed when it is no longer necessary for the purpose for which it was collected.

8. No, the manager is not abiding by the code of ethics of a computer professional. By delivering a system with a known bug that may have catastrophic consequences, it can potentially harm people's lives. To deliver the system on time might be well intended, but in this case as the fault could have catastrophic consequences, the manager are obliged to mitigate the harm as much as possible.